Technological University of the Philippines – Manila
**College of Engineering**
Electronics Engineering Department
Lab4_DataStructures

## 1. Engineering Test Data Organizer

### Generated Test Data (28 Total Records):

**Sources:** Royo_T1 through Royo_T5

**Categories:** TEMPERATURE, VOLTAGE, CURRENT, PRESSURE, SPEED

### Sample Values:

Royo_T1 | TEMPERATURE: 113.88

Royo_T1 | VOLTAGE: 102.52

Royo_T1 | CURRENT: 120.46

Royo_T1 | PRESSURE: 92.93

Royo_T1 | SPEED: 85.22

### Repeated/Distinct Data:

**Repeated Record:** Source: Royo_T1, Category: TEMPERATURE, Value: 113.88

**Distinct Records Count:** 27 records

### Numerical and Categorical Summaries:

**TEMPERATURE:** Count: 6 | Min: 78.43 | Max: 116.38 | Mean: 99.41 | StdDev: 13.91

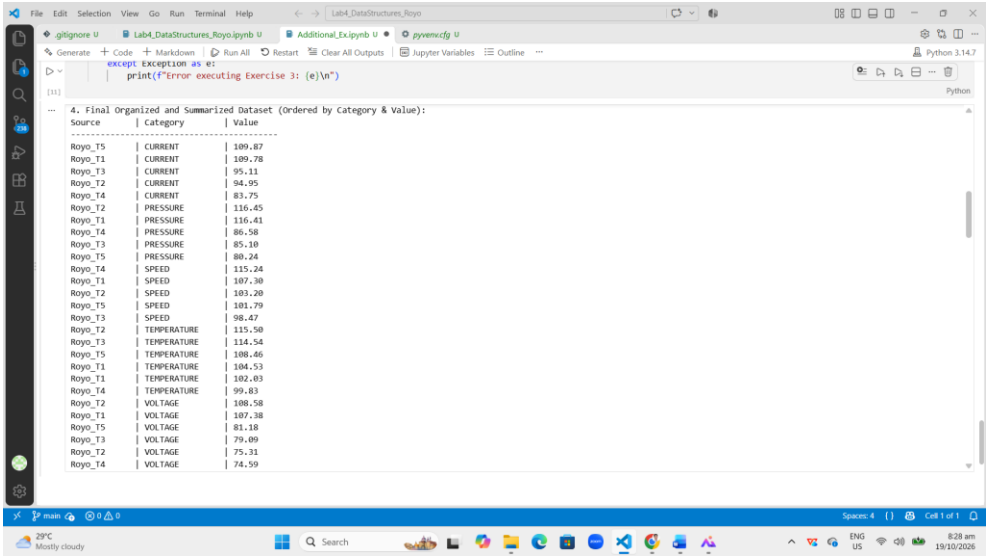
**VOLTAGE:** Count: 6 | Min: 85.50 | Max: 113.33 | Mean: 100.99 | StdDev: 9.07

**CURRENT:** Count: 5 | Min: 81.38 | Max: 120.46 | Mean: 101.99 | StdDev: 13.91

**PRESSURE:** Count: 5 | Min: 80.57 | Max: 113.62 | Mean: 96.09 | StdDev: 12.33

**SPEED:** Count: 5 | Min: 80.49 | Max: 118.88 | Mean: 97.20 | StdDev: 15.65

### Final Organized Dataset (Ordered by Category and Descending Value):



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## 2. Student Performance Data Analyzer

### Generated Performance Records (27 Total Records):

**Students:** Royo_S1 through Royo_S5

**Subjects:** PROGRAMMING, MATHEMATICS, ELECTRONICS, PHYSICS, ENGLISH

### Sample Scores:

Royo_S1 | PROGRAMMING: 95.0 (Excellent)

Royo_S1 | MATHEMATICS: 88.0 (Excellent)

Royo_S1 | ELECTRONICS: 101.0 (Excellent)

Royo_S1 | PHYSICS: 81.0 (Good)

Royo_S1 | ENGLISH: 76.0 (Good)

### Repeated / Incomplete Records:

**Repeated Record:** Student: Royo_S1, Subject: PROGRAMMING, Score: 95.0, Category: Excellent

**Incomplete Record:** Student: Royo_S3, Subject: PHYSICS, Score: None, Category: Incomplete

### Subject Performance Summaries:

**PROGRAMMING:** Count: 5 | Min: 73.0 | Max: 95.0 | Mean: 85.0

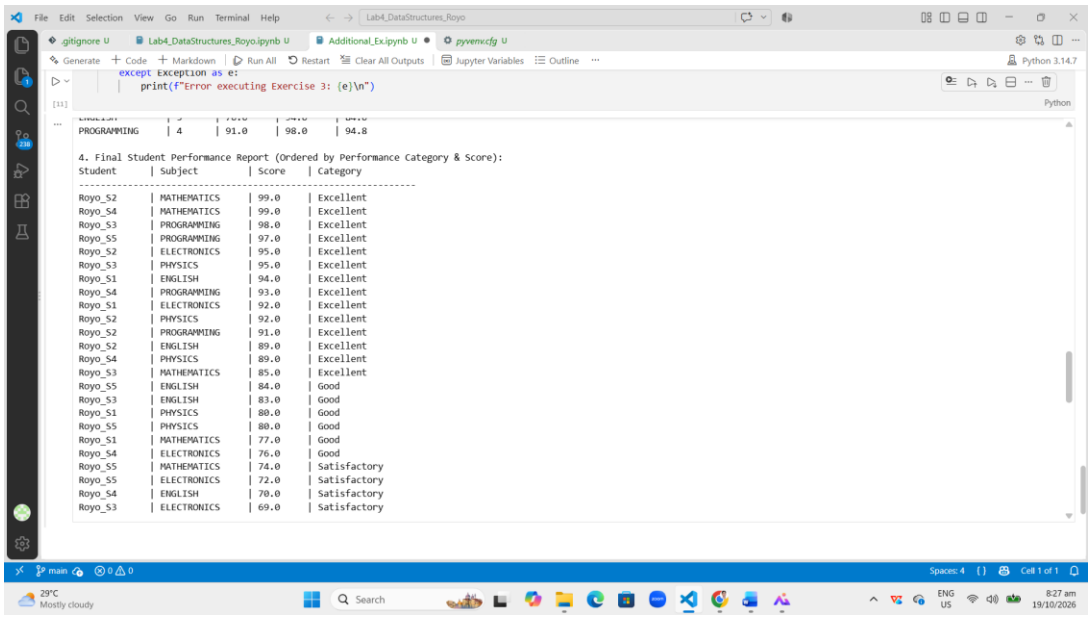
**MATHEMATICS:** Count: 5 | Min: 78.0 | Max: 98.0 | Mean: 88.0

**ELECTRONICS:** Count: 5 | Min: 75.0 | Max: 101.0 | Mean: 88.2

**PHYSICS:** Count: 4 | Min: 74.0 | Max: 98.0 | Mean: 85.0

**ENGLISH:** Count: 5 | Min: 76.0 | Max: 100.0 | Mean: 88.6

### Final Student Performance Report (Ordered by Performance Category and Score):



```
except Exception as e:
    print(f"Error executing Exercise 3: {e}\n")
```

```
PROGRAMMING     | 4     | 91.0     | 98.0     | 94.8

4. Final Student Performance Report (Ordered by Performance Category & Score):
Student      | Subject         | Score    | Category
-------------------------------------------------------------
Royo_S2      | MATHEMATICS     | 99.0     | Excellent
Royo_S4      | MATHEMATICS     | 99.0     | Excellent
Royo_S3      | PROGRAMMING     | 98.0     | Excellent
Royo_S5      | PROGRAMMING     | 97.0     | Excellent
Royo_S2      | ELECTRONICS     | 95.0     | Excellent
Royo_S3      | PHYSICS         | 95.0     | Excellent
Royo_S1      | ENGLISH         | 94.0     | Excellent
Royo_S4      | PROGRAMMING     | 93.0     | Excellent
Royo_S1      | ELECTRONICS     | 92.0     | Excellent
Royo_S2      | PHYSICS         | 92.0     | Excellent
Royo_S2      | PROGRAMMING     | 91.0     | Excellent
Royo_S2      | ENGLISH         | 89.0     | Excellent
Royo_S4      | PHYSICS         | 89.0     | Excellent
Royo_S3      | MATHEMATICS     | 85.0     | Excellent
Royo_S5      | ENGLISH         | 84.0     | Good
Royo_S3      | ENGLISH         | 83.0     | Good
Royo_S1      | PHYSICS         | 80.0     | Good
Royo_S5      | PHYSICS         | 80.0     | Good
Royo_S1      | MATHEMATICS     | 77.0     | Good
Royo_S4      | ELECTRONICS     | 76.0     | Good
Royo_S5      | MATHEMATICS     | 74.0     | Satisfactory
Royo_S5      | ELECTRONICS     | 72.0     | Satisfactory
Royo_S4      | ENGLISH         | 70.0     | Satisfactory
Royo_S3      | ELECTRONICS     | 69.0     | Satisfactory
```

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## 3. Transaction History Analyzer

**Generated Transaction Records (26 Total Records):**

**Customers:** Royo_C1 through Royo_C5

**Products & Prices:** NOTEBOOK ($50), PEN ($20), USB ($300), HEADPHONES ($500), CALCULATOR ($450)

**Repeated Customer-Product Transaction:**

Customer: Royo_C1, Product: NOTEBOOK, Quantity: 6, Price: 50, Total Value: 300

**Customer Spend Summary:**

Royo_C1**:** Total Spent = $3,460.00

Royo_C2**:** Total Spent = $4,120.00

Royo_C3**:** Total Spent = $3,910.00

Royo_C4**:** Total Spent = $3,270.00

Royo_C5**:** Total Spent = $4,120.00

**Product Sales Summary:**

**NOTEBOOK:** 33 Units Sold | Total Revenue = $1,650.00

**PEN:** 22 Units Sold | Total Revenue = $440.00

**USB:** 23 Units Sold | Total Revenue = $6,900.00

**HEADPHONES:** 12 Units Sold | Total Revenue = $6,000.00

**CALCULATOR:** 8 Units Sold | Total Revenue = $3,600.00

**Overall Transaction Summary:**

